

Can Machine Learning Variable Selection and Individual Commodity Fundamentals Improve Futures Return Forecasting?

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Abstract

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1 Introduction

1.1 Motivation & Background

(Guidolin and Pedio, 2021)

The financialization of commodity markets; why forecasting commodity futures matters to investors, hedgers, and risk managers; the tension between statistical and economic loss functions.

1.2 The Research Gap

Three specific limitations of Guidolin & Pedio (2021): fragility of stepwise AIC selection, use of only aggregate commodity factors rather than individual fundamentals, and a sample that ends in 2012 — missing the COVID shock and post-GFC cycles.

1.3 Research Questions & Hypotheses

This thesis aims at answering whether replacing AIC-based stepwise regression with penalized ML methods (LASSO/Elastic Net) improve the out-of-sample forecasting accuracy of commodity futures returns, both statistically and in portfolio returns. Furthermore, it assesses if commodity-specific individual fundamental predictors (inventory levels, crop reports, weather data) add forecasting power beyond aggregate cross sectional factors on an extended 2004-2024 sample.

1.4 Contributions to the Literature

Methodological contribution (ML selection), empirical contribution (individual fundamentals), and out-of-sample robustness contribution (extended sample).

1.5 Thesis Outline

Explain the structure of the thesis.

2 Literature Review

2.1 Theories of Commodity Futures Returns

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Lag	ACF_lower	ACF	ACF_upper
1	0.973131	0.999222	1.025313
2	0.953351	0.998518	1.043686
3	0.939523	0.997812	1.056100
4	0.928182	0.997124	1.066067
5	0.918358	0.996504	1.074650
6	0.909523	0.995888	1.082252
7	0.901448	0.995305	1.089161
8	0.893894	0.994680	1.095466
9	0.886786	0.994047	1.101308
10	0.879995	0.993353	1.106712
11	0.873548	0.992686	1.111823
12	0.867391	0.992032	1.116673
13	0.861472	0.991377	1.121282
14	0.855780	0.990737	1.125694
15	0.850308	0.990128	1.129949

Table 1: ACF comparison (with bounds)

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	PG US Equity
count	5642.000000
mean	0.000210
std	0.011285
min	-0.091428
25%	-0.005063
50%	0.000354
75%	0.005859
max	0.113409
skewness	-0.149761
kurtosis	12.483725
variance	0.000127
median	0.000354
mode	0.000000
sum	1.183665
sum of squared deviations	0.718408
jarque-bera	21122.533522
jarque-bera probability value	0.000000

Table 2: Summary statistics

2.2 Aggregate Commodity-Specific Factors

Model	AIC	BIC	HQIC
CER	-34588.171062	-34581.533068	-34585.858721
ARMA(3,2)	-34626.324616	-34579.862382	-34610.139093
ARMA(5,5)	-34620.963843	-34541.320684	-34593.218710
ARMA(0,5)	-34621.488221	-34575.029712	-34605.303561

Table 3: Model Comparison: Information Criteria

2.3 Individual Commodity Fundamentals

2.4 Macroeconomic Predictors

2.5 ML Methods in Return Forecasting

2.6 Volatility Regimes and Predictability

Series	Test	Statistic	p-value	Conclusion	Suggests
Y_t	ADF	-0.8753	0.7960	Fail to reject H0 (unit root)	non-stationarity
	PP	-1.0501	0.7346	Fail to reject H0 (unit root)	non-stationarity
	KPSS	0.3898	< 0.0816	Fail to reject H0 (stationarity)	stationarity
ΔY_t	ADF	-16.6776	0.0000	Reject H0 (unit root)	stationarity
	PP	-16.8665	0.0000	Reject H0 (unit root)	stationarity
	KPSS	0.2874	< 0.1000	Fail to reject H0 (stationarity)	stationarity
$\Delta^2 Y_t$	KPSS	0.0207	> 0.1000	Fail to reject H0 (stationarity)	stationarity

Table 4: Stationarity Tests on Dividend Series

Model	RMSE	R2
OLS	0.421	0.120
LASSO	0.387	0.180
Elastic Net	0.372	0.210

Table 5: Model comparison

3 Methodology

3.1 Baseline Framework (Replication)

We begin by gathering all T observations of n variables into a data matrix $\mathbf{X} \in \mathbb{R}^{T \times n}$, where each row is a *month* and each column is a macroeconomic variable. Each column is demeaned and standardized to unit variance using the StandardScaler:

$$X_{\text{scaled},ij} = \frac{X_{\text{raw},ij} - \mu_j}{\sigma_j}. \quad (1)$$

Let $\Sigma = V[\mathbf{X}] \in \mathbb{R}^{n \times n}$ denote the variance-covariance matrix of $\mathbf{X}$, which is calculated as:

$$\Sigma = \frac{1}{T-1} \mathbf{X}'\mathbf{X}, \quad (2)$$

and let $\Sigma = \mathbf{U}\mathbf{\Lambda}\mathbf{U}'$ be its spectral decomposition, where $\mathbf{U} \in \mathbb{R}^{n \times n}$ is an orthogonal matrix of order n , whose columns are the eigenvectors $\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n$ of Σ , and $\mathbf{\Lambda} \in \mathbb{R}^{n \times n}$ is a diagonal matrix of the same order, whose diagonal entries are the corresponding eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$, which, by convention, are sorted from the largest to the smallest. By multiplying the original scaled data matrix $\mathbf{X}$ with $\mathbf{U}$, we derive the principal components of $\mathbf{X}$ as the elements of the matrix $\mathbf{F} \in \mathbb{R}^{T \times n}$, or formally:

$$\mathbf{F} = \mathbf{X}\mathbf{U}. \quad (3)$$

To assess how much of the total variance in the original n variables is captured by the k -th principal component, or alternatively by the first K principal components, we can compute the variance-covariance matrix of $\mathbf{F}$ as:

$$\begin{aligned} V[\mathbf{F}] &= V[\mathbf{X}\mathbf{U}] \\ &= \mathbf{U}'\Sigma\mathbf{U} \\ &= \mathbf{U}'\mathbf{U}\mathbf{\Lambda}\mathbf{U}'\mathbf{U} \\ &= \mathbf{I}_n\mathbf{\Lambda}\mathbf{I}_n \\ &= \mathbf{\Lambda}, \end{aligned} \quad (4)$$

where we exploited the orthogonality of $\mathbf{U}$. Recalling the fact that $\mathbf{\Lambda}$ is a diagonal matrix, $V[\mathbf{F}] = \mathbf{\Lambda}$ implies that the principal components are mutually uncorrelated, and have a variance equal to the associated eigenvalue. Therefore, the proportion of variance in the original n variables explained by the k -th principal component is given by:

$$\frac{\lambda_k}{\sum_{i=1}^n \lambda_i}, \quad (5)$$

and the cumulative proportion of explained variance by the first K components by:

$$\frac{\sum_{k=1}^K \lambda_k}{\sum_{i=1}^n \lambda_i}. \quad (6)$$

Mirroring the approach used by Guidolin and Pedio, 2021, we forecast the returns of each commodity j , $R_{t+1,j}$, with the principal component analysis to summarize a large set of macroeconomic factors:

$$R_{t+1,j} = \underbrace{\alpha_j}_{\text{intercept}} + \underbrace{\sum_{i=1}^N \beta_{ij} PC_{t,ij}}_{\text{macroeconomic}} + \underbrace{D_{t,j} \left(\sum_{k=1}^K \gamma_{kj} C_{t,kj} \right)}_{\text{commodity-specific}} + \underbrace{\varepsilon_{t,j}}_{\text{error term}} \quad (7)$$

where α_j is the model's intercept,

3.2 Extension I: Penalized Regression

In the first alternate approach, we keep the principal component analysis and still extract ten principal components from the n initial variables, but instead of running stepwise AIC on those selected principal components, we run a LASSO regression as an alternative way to further reduce the number of parameters to be estimated. With this only being a narrow replacement compared to the original framework, every difference in results should be attributable solely to the selection method.

$$R_{t+1,j} = \alpha_j + \sum_{i=1}^{10} \beta_{ij}^{\text{LASSO}} PC_{t,ij} + \sum_{k=1}^K \gamma_{kj}^{\text{LASSO}} C_{t,kj} + \varepsilon_{t,j} \quad (8)$$

A broader extension to the approach by Guidolin and Pedio, 2021 is foregoing the principal component analysis entirely and applying LASSO directly to the n variables.

$$R_{t+1,j} = \alpha_j + \sum_{i=1}^{128} \beta_{ij}^{\text{LASSO}} X_{t,i} + \sum_{k=1}^K \gamma_{kj}^{\text{LASSO}} C_{t,kj} + \varepsilon_{t,j} \quad (9)$$

3.3 Extension II: Individual Fundamental Predictors

3.4 OOS Evaluation Criteria

3.5 Portfolio Back-Testing Framework

4 Data

- 4.1 Commodity Futures Returns**
- 4.2 Macroeconomic Factors**
- 4.3 Aggregate Commodity-Specific Factors**
- 4.4 Individual Fundamental Predictors**
- 4.5 Sample Splits and Descriptive Statistics**

5 Replication & Baseline Results

5.1 In-Sample Estimation (1989-2003)

5.2 OOS Predictive Accuracy (2004-2012)

5.3 Portfolio Back-Testing (2004-2012)

5.4 Replication Assessment

6 Extension I: ML Variable Selection

6.1 Model Specification and Implementation

6.2 Variable Selection Patterns

6.3 OOS Forecasting Results - Original Sample (2004-2012)

6.4 OOS Forecasting Results - Extended Sample (2013-2024)

6.5 Portfolio Back-Testing

6.6 Robustness Checks

7 Extension II: Individual Commodity Fundamentals

7.1 Predictor Construction by Commodity Group

7.2 Individual vs. Aggregate Factors: Are They Substitutes?

7.3 OOS Forecasting Results (2004-2024)

7.4 Combined Model

7.5 Portfolio Back-Testing

7.6 Regime Interaction

8 Synthesis & Discussion

8.1 Model Horse Race

8.2 Statistical vs. Economic Loss Functions Revisited

8.3 Implications for Commodity Market Segmentation

8.4 Practical Implications

9 Conclusions

9.1 Summary of Findings

9.2 Limitations

9.3 Future Research Directions

References

Guidolin, Massimo and Manuela Pedio (Apr. 2021). “Forecasting Commodity Futures Returns with Stepwise Regressions: Do Commodity-Specific Factors Help?” In: *Annals of Operations Research* 299.1–2, pp. 1317–1356. ISSN: 0254-5330, 1572-9338. DOI: 10.1007/s10479-020-03515-w. URL: <http://link.springer.com/10.1007/s10479-020-03515-w> (visited on 04/17/2026).

Appendix